Silver Data Quality Checks

This document defines a small set of checks to run after building the Lite Silver tables. These checks validate joinability, basic referential integrity and expected value ranges.

5 Main Checks

1. Dimension uniqueness: SKU unique in silver__product_master__lite; location_id unique in silver__location_master__lite.
2. Key coverage: sales/snapshots/movements SKUs and locations should exist in product and location dimensions.
3. Date parsing success: transaction_date, snapshot_date, movement_date should be populated (review nulls).
4. Reasonable ranges: qty and on_hand_qty within plausible limits (review extreme negatives/outliers).
5. Movement type validity: movement_type limited to SALE, REPLENISH, ADJUSTMENT after normalization.
6.

Example SQL Snippets (DuckDB)

Run these directly in DuckDB to validate the Lite Silver layer.

1) SKU uniqueness

```
SELECT sku, COUNT(*) AS cnt
FROM silver__product_master__lite
GROUP BY 1
HAVING COUNT(*) > 1;
```

2) Location uniqueness

```
SELECT location_id, COUNT(*) AS cnt
FROM silver__location_master__lite
GROUP BY 1
HAVING COUNT(*) > 1;
```

3) Fact SKUs not in product dimension

```
SELECT DISTINCT s.sku
FROM silver__sales_transactions__lite s
LEFT JOIN silver__product_master__lite p ON s.sku = p.sku
WHERE p.sku IS NULL
LIMIT 100;
```

4) Date null rate (sales)

```
SELECT COUNT(*) AS total_rows,
       SUM(CASE WHEN transaction_date IS NULL THEN 1 ELSE 0 END) AS null_dates
FROM silver__sales_transactions__lite;
```

5) Movement type validity

```
SELECT movement_type, COUNT(*) AS cnt
FROM silver__inventory_movements__lite
GROUP BY 1
ORDER BY cnt DESC;
```